Lecture 15: Bases for row(A), col(A) and null(A).

Def: If A is m by n, row(A) is the subspace of $\mathbb{R}^n$ spanned by the rows of A.

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{U}$$

Fact 1: $\text{row}(A) = \text{row}(U)$

$$\dim \text{row}(A) = \dim \text{row}(U) = \#\text{ pivots} = r$$

Fact 2: $\text{null}(A) = \text{null}(U)$

$$\dim \text{null}(A) = \dim \text{null}(U) = \#\text{ free vars} = n - r$$

Example:

$$x_1 + 3x_2 + 3x_3 + 2x_4 = 0$$
$$3x_3 + 3x_4 = 0$$

Free: x_2, x_4

- $x_2 = 1,\ x_4 = 0$:
$$\begin{cases} x_1 + 3x_3 = -3 \\ 3x_3 = 0 \end{cases}$$
- $x_2 = 0,\ x_4 = 1$:
$$\begin{cases} x_1 + 3x_3 = -2 \\ 3x_3 = -3 \end{cases}$$

$\text{null}(A) = \text{null}(U)$:

$$s_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + s_2 \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

Fact 3: $\text{col}(A) \neq \text{col}(U)$, <u>but</u> $\dim \text{col}(A) = \dim \text{col}(U)$.
(same number of indep. cols)

Reason: $Ax = 0$ exactly when $Ux = 0$.

The relation
$$\text{col } 4 = \text{col } 3 - \text{col } 1$$
is true for U, hence for A.

Pivot columns (1 and 3) are lin. indep. in U, so also in A.

$$\dim \operatorname{col}(A) = \dim \operatorname{col}(U)$$
$$= \#\text{ pivots} = r.$$

<u>Theorem</u>: For any m by n matrix A,

1) $\dim \operatorname{row}(A) = \dim \operatorname{col}(A) = r$
2) $\underbrace{\dim \operatorname{null}(A)}_{\text{nullity}(A} = n - \dim \operatorname{col}(A)$

number of columns.

call this rank(A)